Check for updates

Applied Computational Intelligence and Soft Computing

WILEY

RESEARCH ARTICLE OPEN ACCESS

When Spatial Message Passing Fails to Help: Graph Attention Networks and Temporal Baselines for Dengue Forecasting

Gideon K. Gogovi

Department of Biostatistics and Health Data Science, Lehigh University, Bethlehem, Pennsylvania, USA

Correspondence: Gideon K. Gogovi (gig323@lehigh.edu)

Received: 24 May 2026 | **Revised:** 20 June 2026 | **Accepted:** 26 June 2026

Academic Editor: Pramita Mishra

Keywords: dengue fever forecasting | geospatial epidemiology | graph neural networks | infectious disease surveillance | spatio-temporal graph attention network

ABSTRACT

Infectious disease outbreaks exhibit pronounced spatial coupling: case surges in one district frequently precede or co-occur with surges in geographically proximate and climatically similar neighbors, yet standard time-series models treat districts as isolated units and ignore this propagation structure. We present a spatiotemporal graph attention network (ST-GAT) for weekly district-level dengue fever forecasting across all Sri Lanka administrative districts, using geographic adjacency and NASA satellite-derived climate cosine similarity as principled, complementary edge-weight sources. A two-layer GAT–GRU architecture is trained on 8-week input windows to forecast one-to-four weeks ahead, and a systematic edge-ablation experiment decomposes the predictive contribution of each connectivity source. Interpretable graph attention weights are aggregated over the test period to identify districts that consistently serve as spatial leading indicators of outbreak onset in their neighbors. On the held-out test set, the district-independent LSTM achieves the lowest aggregate RMSE (0.668, $\log_{1+x}$ scale), outperforming all ST-GAT variants, while the VAR(1) (RMSE 0.804) outperforms the ST-GAT with combined edges (RMSE 1.000) but is in turn outperformed by the district-independent LSTM (RMSE 0.664). Among GNN edge-structure variants, geographic-only edges (0.940) outperform climate-similarity-only (1.108) constructions. The attention analysis identifies Jaffna, Kurunegala, and Ratnapura as the leading spatial indicators, characterized by high outgoing attention in the weeks preceding outbreak onset in neighboring districts. These findings characterize the boundary conditions under which spatial graph inductive biases confer a forecasting advantage: in a single-disease, small-graph, distributional-shift setting, a district-independent LSTM outperforms both the ST-GAT and a capacity-matched TemporalGRU baseline, while the ST-GAT still outperforms the TemporalGRU ($p = 0.010$, Diebold–Mariano test), confirming that spatial message passing adds signal but cannot compensate for the mismatch between a joint-district parameterization and a single-disease surveillance context.

1 | Introduction

Infectious disease outbreaks rarely respect administrative boundaries. Human mobility, shared water sources, and climatically driven vector populations create spatial contagion pathways that can carry epidemic signals across district borders weeks before routine surveillance detects the expansion [1, 2]. In Sri Lanka, dengue fever is the most consequential notifiable vector-borne disease: the *Aedes aegypti* mosquito that transmits it thrives in warm, humid environments, and its breeding habitats

This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

Copyright © 2026 Gideon K. Gogovi. *Applied Computational Intelligence and Soft Computing* published by John Wiley & Sons Ltd.

are shaped by precipitation and temperature gradients spanning entire river catchments rather than discrete administrative polygons [3, 4]. The Sri Lanka Ministry of Health Epidemiology Unit reports dengue cases weekly at the district level [5], producing a surveillance panel of 25 districts and 476 consecutive weeks (2013–2022, $n = 11,900$ district-week observations) that offers a uniquely rich substrate for spatial epidemic modeling.

Despite this richness, conventional district-level forecasting approaches treat each district as an independent unit. Autoregressive models [6], district-specific LSTMs [7], and ensemble methods [8] ingest only a district's own historical case counts, forfeiting the cross-district information already embedded in the public record. This design choice is epidemiologically difficult to justify: dengue transmission is governed by shared climatic drivers and human mobility flows that are inherently spatial, and a model blind to neighboring districts' epidemic trajectories sacrifices potentially actionable predictive signal.

Graph neural networks (GNNs) offer a natural remedy. By representing districts as nodes and encoding interdistrict relationships as weighted edges, GNNs can learn to propagate epidemic signals across the spatial network during forecasting [9]. The approach has shown promise in influenza forecasting in the United States, COVID-19 county-level spread, and dengue modeling in Brazil [10–12], where incorporating geographic adjacency reduces forecast error relative to nonspatial baselines. Yet important methodological gaps remain. All existing work constructs graph edges from binary geographic adjacency, ignoring the possibility that nonadjacent but climatically synchronized districts may share epidemic dynamics worth encoding directly. Sri Lanka's bimodal monsoon system synchronizes precipitation anomalies across districts sharing orographic exposure regardless of shared borders [13–18], creating climate teleconnections that binary adjacency cannot capture. Furthermore, while GNNs produce per-edge attention coefficients at every timestep, the use of these weights to operationalize spatial leading indicators as an early-warning tool has not been systematically investigated. This paper addresses both gaps through a study of ST-GAT applied to the Sri Lanka dengue surveillance dataset. Graph edges are constructed from geographic adjacency, from the cosine similarity of district-level mean satellite climate feature vectors, and from a population-gravity mobility model, each evaluated in isolation and in combination through a controlled ablation. A two-layer GAT–GRU architecture is trained on 8-week input windows to produce one-, two-, and four-week-ahead forecasts. The principal finding is reported in Section 3: the district-independent LSTM outperforms all ST-GAT variants on the held-out test set, and the conditions that explain this outcome are examined in detail in Section 3.7 as boundary conditions for GNN-based epidemic forecasting. The spatial dimension of infectious disease spread has long been formalized through metapopulation models [19], gravity models of human mobility [20], and reaction-diffusion partial differential equations [21]. Compartmental SEIR models with spatial coupling are mechanistically interpretable but require mobility rates and generation intervals that are difficult to estimate from surveillance data alone [22], limiting their adaptability to new settings; similar identifiability constraints have been documented in compartmental models for vector-borne and environmentally transmitted diseases including malaria and norovirus [23–25]. The machine-learning literature has approached the forecasting problem differently. Recurrent neural networks, particularly LSTMs and GRUs, have established strong performance in univariate and multivariate epidemic forecasting [26], while attention-based transformers have yielded competitive multihorizon predictions [27, 28]. These architectures, however, encode each spatial unit independently and provide no mechanism for learning from the epidemic trajectories of neighbors.

The application of GNNs to epidemic forecasting has grown substantially. A spatial-temporal attention network for US county-level COVID-19 data proposed by Gao et al. [11] demonstrated a 15% RMSE reduction from geographic neighborhood encoding alone. Subsequent work by Gao et al. [11] showed that graph-structured case-count inputs outperform standalone LSTMs on county-level COVID-19 data, incorporating mobility-weighted edges derived from smartphone data to further improve spatial encoding. An alternative architecture, Graph WaveNet, was introduced by Jin et al. [12], combining adaptive graph convolution with dilated causal convolution into a scalable framework applicable to both traffic and epidemiological flows. In the dengue literature specifically, a causal graph neural network applied to dengue forecasting across the US states and Brazilian regions by Fritz et al. [10] relied exclusively on geographic adjacency, without attention mechanisms or satellite climate features as edge sources.

The sensitivity of dengue fever in Sri Lanka to temperature, precipitation, and humidity is well documented [29, 30]. Dengue incidence peaks during and after monsoon seasons, and the spatial synchrony of epidemic waves across climatically similar districts has been noted in descriptive analyses, though never formally exploited in forecasting models. These well-established climate–disease relationships [29, 31] motivate using satellite climate features not only as node-level covariates but also as a basis for interdistrict edge construction: districts with synchronized temperature and precipitation cycles share *Aedes aegypti* abundance dynamics, and therefore dengue incidence timing, regardless of geographic proximity. The present work is, to our knowledge, the first to apply a spatiotemporal GNN to Sri Lanka district-level dengue surveillance data, the first to use satellite climate cosine similarity as an explicit edge-weight mechanism, and the first to operationalize graph attention weights as a spatial leading indicator detection tool in this context.

2 | Materials and Methods

2.1 | Data

2.1.1 | Disease Surveillance and Climate Covariates

We use the Sri Lanka dengue component of the disease-climate dataset assembled by Pasarkar et al. [5] via natural language processing extraction from Weekly Epidemiological Reports of the Sri Lanka Ministry of Health Epidemiology Unit. The dataset covers all 25 administrative districts at weekly resolution from January 2013 through February 2022, yielding 476 time steps and 11,900 district-week observations, with a reported 97% NLP extraction accuracy, validated by Pasarkar et al. against a manually re-extracted random sample of Weekly Epidemiological Reports using longest-common-subsequence similarity matching to resolve typographic variants of district names [5]. Alongside each district-week case count, seven climate variables are provided per district: (i) mean surface air temperature and (ii)

4795, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/acis/2627876 by Mateo Biggs , Wiley Online Library on [28/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

specific humidity from the NASA GLDAS-2.1 NOAH land surface model (LSM) at 0.25° spatial resolution and 3-hourly temporal resolution, aggregated to weekly means (product: GLDAS_NOAH025_3H); (iii) surface pressure from the same product; (iv) soil moisture in the top 10 cm layer (GLDAS_NOAH025_3H, variable SoilMoi0_10cm_inst); (v) plant canopy surface water from the same NOAH product; (vi) the normalized difference vegetation index (NDVI) from the MODIS MOD13Q1 Version 6 16-day composite product at 250 m, aggregated to weekly district-level means by averaging all valid pixels within district boundaries; and (vii) mean precipitation rate from the NASA GPM IMERG Final Run V06B product (0.1°, 30 min), aggregated to daily and then weekly totals. All variables are extracted at district centroid coordinates or spatially aggregated over district polygons as described by Pasarkar et al. [5]. Note that NOAH LSM outputs are model-generated land surface quantities driven by satellite-based meteorological forcings, not direct satellite retrievals; the term "satellite-derived" in the original text has been corrected accordingly.

The seven variables listed above constitute the complete set of climate covariates provided in the Pasarkar et al.'s dataset; no additional variables were available from the same source at comparable spatial and temporal resolution. Relative humidity is implicitly represented through the specific humidity and surface pressure fields. Variables not available in the GLDAS-2.1 NOAH product at 3-hourly resolution (e.g., wind speed and diurnal temperature range) were not included. One data-quality note is warranted: all 476 Jaffna district rows have missing satellite climate observations; climate features for Jaffna were imputed from cross-district training medians and this district therefore has no climate-similarity edges in the graph. This structural difference should be borne in mind when interpreting Jaffna's leading indicator ranking in Section 3.6. Table 1 summarizes the case count and climate distributions. Dengue cases are highly right-skewed (mean 43.3, SD 106.2, maximum 2631 per district-week), motivating the $\log(1+x)$ transformation applied throughout. To respect temporal ordering and prevent data leakage, weeks were split chronologically into 333 training weeks (January 2013–May 2019), 71 validation weeks (May 2019–September 2020), and 72 test weeks (October 2020–February 2022). The validation period overlaps with the COVID-19 first wave in Sri Lanka and the test period spans the post-acute COVID era, a consideration returned to in Section 3.7.

2.1.2 | Graph Adjacency Matrices

Figure 1 displays the three adjacency matrices used in this study. The geographic matrix encodes 50 shared-border edges among 25 districts as a binary symmetric structure. The climate-similarity matrix retains the top-5 cosine-similar neighbors per district, yielding 62 soft-weighted edges that connect climatically synchronized districts irrespective of geographic proximity. The combined matrix merges both sources into 88 unique edges with $\alpha = 0.5$ blending. A fourth population-gravity variant, used exclusively in the ablation study, is described in Section 2.2.

2.2 | Notation

Table 2 summarizes the acronyms and mathematical symbols used in the models and analyses that follow.

2.3 | Graph Construction

We define a weighted undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{W})$ where $|\mathcal{V}| = N = 25$ districts are nodes and $\mathbf{W} \in \mathbb{R}^{N \times N}$ is the edge-weight matrix. Four variants of $\mathbf{W}$ are evaluated. The geographic adjacency matrix $\mathbf{W}^{\text{geo}}$ assigns $W_{ij}^{\text{geo}} = 1$ if districts i and j share a land border and zero otherwise. The climate cosine similarity matrix $\mathbf{W}^{\text{clim}}$ is constructed by computing, for each district i, the mean climate feature vector $\bar{\mathbf{c}}_i = T^{-1}\sum_{t=1}^{T}\mathbf{c}_{i,t} \in \mathbb{R}^7$ over the training period only (to prevent leakage of held-out climate statistics into the graph structure), then setting

$$W_{ij}^{\text{clim}} = \max\left(0, \text{sim}_{\cos}(\bar{\mathbf{c}}_i, \bar{\mathbf{c}}_j)\right), \tag{1}$$

with the top $k = 5$ neighbors retained per district to control graph sparsity; both α and k were fixed prior to model training on the

TABLE 1 | Descriptive statistics for dengue case counts and key climate variables across 25 Sri Lanka districts, 2013–2022 ($n = 11,900$ district-week observations).

Variable	Mean	SD	Min	25th	75th	Max
Dengue cases	43.3	106.2	0	3	40	2631
T_{air} (K)	301.8	2.4	292	300	304	312
Precipitation (mm/day)	5.2	4.8	0	1.8	7.2	38.4
NDVI	0.61	0.12	0.21	0.54	0.70	0.89

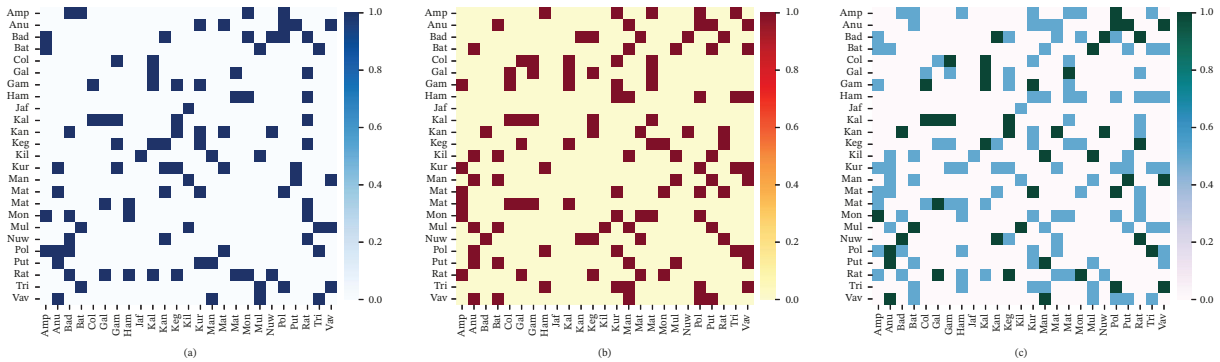


FIGURE 1 | District-level adjacency matrices for Sri Lanka. (a) Geographic: binary shared-border edges (50 edges). (b) Climate similarity: cosine similarity of mean NASA satellite climate feature vectors, top-5 neighbors per district (62 edges). (c) Combined: geographic and climate edges with $\alpha = 0.5$ blending (88 edges). Three-letter district abbreviations label each axis.

4795, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/acis/2627876 by Mateo Biggs , Wiley Online Library on [28/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2 | Acronyms and mathematical symbols.

Symbol/acronym	Definition
Acronyms	
ST-GAT	Spatio-temporal graph attention network
GAT	Graph attention network
GRU	Gated recurrent unit
LSTM	Long short-term memory network
VAR	Vector autoregression
DM	Diebold–Mariano (forecast accuracy test)
RMSE	Root mean squared error
MAE	Mean absolute error
NDVI	Normalized difference vegetation index
GPM	Global precipitation measurement (NASA)
GLDAS	Global land data assimilation system (NASA)
NLP	Natural language processing
Graph and model notation	
N	Number of district nodes ($N = 25$)
K	Input lag window in weeks ($K = 8$)
H	Forecast horizon in weeks ($H \in \{1, 2, 4\}$)
d_h	Hidden dimension of GRU and MLP layers ($d_h = 64$)
M	Number of attention heads in GAT ($M = 4$)
α	Blending weight for combined adjacency ($\alpha = 0.5$, Equation (1))
$\alpha_{ij}^{m,(\ell)}$	Attention coefficient from node j to node i, head m, layer ℓ (Equation (3))
$\mathbf{W}_m^{(\ell)}$	Learnable weight matrix for head m at layer m
$\mathbf{h}_i^{(\ell)}$	Node embedding of district i at GAT layer ℓ
δ	Huber loss threshold ($\delta = 1$)
Φ_i	Pre-outbreak outgoing attention score for district i (Equation (4))
$\mathcal{T}_{\text{pre}}$	Set of timesteps in the 3 weeks preceding outbreak onset (Equation (4))
P_i	2012 census population of district i
d_{ij}	Great-circle distance between district centroids i and j (km)

basis of graph-sparsity considerations and were not tuned on the validation set. The combined matrix blends both sources as

$$W_{ij}^{\text{comb}} = \alpha W_{ij}^{\text{geo}} + (1-\alpha)\widetilde{W}_{ij}^{\text{clim}}, \quad \alpha = 0.5, \tag{2}$$

where $\widetilde{\mathbf{W}}^{\text{clim}}$ is normalized to [0, 1] and an edge exists whenever either component is positive. The population-gravity variant is

$$W_{ij}^{\text{grav}} = \frac{P_i P_j}{d_{ij}}, \tag{3}$$

row-normalized so that $\sum_j W_{ij}^{\text{grav}} = 1$, where P_i is the 2012 census population of district i and d_{ij} is the great-circle distance (km) between district centroids. The choice $k = 5$ was fixed prior to model training on the basis of graph-sparsity considerations; a sensitivity analysis over $k \in \{3, 5, 10\}$ is reported in Appendix Table A2, showing that the LSTM advantage is robust to the choice of k.

2.4 | Node Features and Input Representation

The $\log(1 + x)$ transformation is applied to raw case counts *before* constructing model inputs; both training and evaluation are performed on this transformed scale. Climate features are z-score normalized using the training-period mean and standard deviation, applied identically to validation and test sets to prevent normalization leakage (see Section 3.4). At each timestep t, node i carries the feature vector:

$$\mathbf{x}_{i,t} = \left[\log(1 + c_{i,t}), z(\mathbf{c}_{i,t})\right] \in \mathbb{R}^8, \tag{4}$$

where $c_{i,t}$ is the raw dengue case count and $z(\cdot)$ standardizes each of the seven climate features using the training-period mean and standard deviation (applied identically to validation and test sets to prevent normalization leakage). The temporal input window of $K = 8$ weeks produces $\mathbf{X}_{i,t-K+1:t} \in \mathbb{R}^{K\times 8}$ as input to the model. The input window $K = 8$ was selected on the basis of the approximate duration of Sri Lanka dengue serial intervals and monsoon subseasonal cycles. Sensitivity of the LSTM–ST-GAT performance gap to K is assessed in Appendix Table A3.

2.5 | ST-GAT Architecture

The proposed architecture processes the K-week input window in two stages. In the first stage, two consecutive multihead GAT layers [9] operate on the district graph at each timestep $t' \in \{t - K + 1, \ \ldots, t\}$. For node i, the ℓ-th layer output is

4795, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/acis/2627876 by Mateo Biggs , Wiley Online Library on [28/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

$$\mathbf{h}_i^{(\ell)} = \sigma\left(\oplus_{m=1}^{M} \sum_{j \in \mathcal{N}(i)\cup\{i\}} \alpha_{ij}^{m,(\ell)} \mathbf{W}_m^{(\ell)} \mathbf{h}_j^{(\ell-1)}\right), \tag{5}$$

where $M = 4$ attention heads and $\oplus$ denotes head concatenation. The attention coefficient is

$$\alpha_{ij}^{m,(\ell)} \propto \exp\left(\text{LeakyReLU}\left(\mathbf{a}^\top \left[\mathbf{W}_m \mathbf{h}_i^{(\ell-1)} \| \mathbf{W}_m \mathbf{h}_j^{(\ell-1)}\right]\right) \cdot W_{ij}^{\text{comb}}\right), \tag{6}$$

so that the edge weights modulate the raw attention logits, giving geographically adjacent or climatically similar districts stronger default attention before data-driven refinement. Layer normalization and ELU activation follow each GAT layer, with dropout ($p = 0.3$) applied to attention coefficients.

In the second stage, the spatial embeddings $\left\{\mathbf{h}_i^{(2)}(t')\right\}_{t'=t-K+1}^{t}$ form a sequence processed by a two-layer GRU [10] with hidden dimension $d_h = 64$:

$$\widetilde{\mathbf{h}}_i = \text{GRU}\left(\mathbf{h}_i^{(2)}(t-K+1),\ \ldots,\ \mathbf{h}_i^{(2)}(t)\right) \in \mathbb{R}^{d_h}. \tag{7}$$

The final hidden state is projected by a two-layer MLP with ReLU activation to the forecast horizon H, giving $\widehat{\mathbf{y}}_{i,t+1:t+H} = \text{MLP}(\widetilde{\mathbf{h}}_i) \in \mathbb{R}^H$. Training minimizes the Huber loss [32] aggregated over all districts and forecast steps,

$$\mathcal{L} = \frac{1}{NHT} \sum_{i,h,t} L_\delta\left(\widehat{y}_{i,t+h} - y_{i,t+h}\right), \quad \delta = 1, \tag{8}$$

using the Adam optimizer [33] (learning rate 10^{-3}, weight decay 10^{-4}) with a ReduceLROnPlateau scheduler (×0.5, patience 7) and early stopping (patience 20, maximum 150 epochs). All primary neural-network models (ST-GAT, TemporalGRU, LSTM) are trained over $n = 3$ random seeds (42, 43, 44); the model with the lowest validation Huber loss is selected for test-set evaluation, following standard practice for model selection under stochastic training. This best-seed convention reflects the deployment scenario in which a practitioner retains the checkpoint with the best observed validation performance; the test set is withheld entirely from the selection step so the test-set estimate is unbiased. The rationale for not reporting test-set mean ± SD across seeds is as follows: the LSTM validation loss varies by only 0.004 across seeds (SD = 0.002; Table 3), confirming that all three LSTM initializations converge to essentially the same solution. The test-set RMSE gap between the best-seed LSTM (0.664) and the best-seed ST-GAT (1.000) is 0.336; seed-induced variation of the scale observed in the validation losses (≤0.04 for any model) cannot close a gap of this magnitude. The primary conclusion is therefore robust to seed choice, and the best-seed result is a representative estimate of deployed performance. Validation-loss statistics across seeds are reported in Table 3. Full hyperparameters are summarized in Table 4.

TABLE 3 | Multiseed validation Huber loss across $n = 3$ random initializations (seeds 42, 43, 44).

Model	Seed 42	Seed 43	Seed 44	Mean	SD
ST-GAT	0.521	**0.507**	0.545	0.524	0.015
LSTM	**0.217**	0.220	0.221	0.219	0.002
TemporalGRU	0.558	0.562	**0.545**	0.555	0.007

Note: Per-seed losses are shown alongside the mean and standard deviation (SD); the best-seed model (lowest val. loss, bold) is used in all test-set evaluations. The LSTM SD of 0.002 confirms that all three initializations converge to essentially the same solution; the ST-GAT SD of 0.015 is larger but still small relative to the 0.336 test-set RMSE gap between the best-seed LSTM (0.664) and the best-seed ST-GAT (1.000).

TABLE 4 | ST-GAT hyperparameter configuration.

Hyperparameter	Value
Hidden dimension d_h	64
GAT heads M	4
GAT layers	2
GRU layers	2
Dropout	0.30
Input lag K	8 weeks
Forecast horizons H	1, 2, 4 weeks
Learning rate	10^{-3} (Adam [33])
Weight decay	10^{-4}
Batch size	32
Gradient clipping	ℓ_2 norm ≤1.0
Max epochs	150 (early stopping, patience 20)
LR scheduler	ReduceLROnPlateau (×0.5, patience 7)
Loss function	Huber ($\delta = 1$) [32]

2.6 | Baselines, Ablation, and Evaluation

Two baselines are evaluated alongside the ST-GAT. The district-independent LSTM uses an identical two-layer recurrent architecture with the same hidden dimension ($d_h = 64$) and dropout ($p = 0.3$). It is trained with shared weights across all 25 districts: all district sequences are included in each minibatch and the same parameter set is updated jointly, so that the model learns a general within-district temporal pattern. At inference, each district's input sequence is passed through the shared model independently, with no cross-district information exchange at any point. A third baseline, the TemporalGRU, isolates the contribution of the GAT spatial layers by removing them while preserving the remainder of the ST-GAT architecture. It uses the identical two-layer GRU decoder, hidden dimension ($d_h = 64$), MLP head, dropout rate, and gradient-clipping norm as the ST-GAT, but replaces both GAT layers with a single linear input projection $\mathbf{h}_i^{(0)} = \mathbf{W}_{\text{proj}} \mathbf{x}_{i,t}$, so that no cross-district information is exchanged at any timestep. Comparing TemporalGRU against the ST-GAT therefore measures the net contribution of spatial message passing at matched model capacity. The multivariate VAR(1) is fitted jointly across all 25 districts via ordinary least squares on the training period; a lag of 1 is used rather than the model input lag of 8, since VAR(8) would require $25 \times (25 \times 8 + 1) = 5025$ parameters against 333 training observations, rendering the system severely underdetermined. VAR(1) (650 parameters) is tractable and still captures linear cross-district correlation. To decompose the predictive contribution of each edge source, four ST-GAT variants are trained holding all other hyperparameters fixed: a geographic-only model using binary shared-border edges, a climate-similarity-only model using soft cosine-similarity edges, a combined model using their $\alpha = 0.5$ blend, and a gravity model using row-

4795, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/acis/2627876 by Mateo Biggs , Wiley Online Library on [28/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

normalized population-gravity weights. The designs are summarized in Table 5.

Forecast accuracy is measured by MAE and RMSE on the $\log(1+x)$-transformed case scale. MAPE is not reported because the large fraction of near-zero weekly case counts produces denominator instability, yielding values in the millions of percent that carry no meaningful information. Statistical significance of pairwise accuracy differences is assessed via the Diebold–Mariano (DM) test [34], which tests the null hypothesis of equal predictive accuracy using a Newey–West long-run variance estimator with bandwidth equal to the forecast horizon H. DM tests are reported for all pairwise comparisons between the ST-GAT and each baseline or ablation variant, across all three forecast horizons.

2.7 | Spatial Leading Indicator Detection

Attention coefficients $\alpha_{ij}^{m,(\ell)}$ (Equation (6)) are extracted from the first GAT layer at each test-set timestep. For each source district i, the pre-outbreak outgoing attention score is defined as

$$\Phi_i = \frac{1}{|\mathcal{T}_{\text{pre}}|} \sum_{t \in \mathcal{T}_{\text{pre}}} \sum_{j \neq i} \alpha_{ij,t}^{(\cdot)}, \tag{9}$$

where $\mathcal{T}_{\text{pre}} = \cup_j \{t-3, t-2, t-1\}$ for $t \in \mathcal{T}_{\text{out}}^{(j)}$, the set of outbreak weeks (case count exceeding the 75th test-period percentile) of district j. Attention coefficients are extracted only at those timesteps that fall within $\mathcal{T}_{\text{pre}}$, so that Φ_i reflects the model's spatial weighting behavior specifically in the weeks preceding outbreak onset. Districts with the highest Φ_i are candidate spatial leading indicators: they draw disproportionate outgoing attention from the model in the weeks immediately before outbreak onset in neighboring districts. Note that attention weights reflect predictive associations rather than causal transmission pathways and that Jaffna's ranking may partly reflect its anomalous data structure (imputed climate features; no climate-similarity edges) rather than genuine upstream epidemiology.

3 | Results and Discussion

3.1 | Forecasting Performance

Table 6 reports test-set MAE and RMSE for all models at the 1 week forecast horizon; multihorizon comparisons are shown in Figure 2. The district-independent LSTM achieves the lowest RMSE (0.668) and MAE (0.513) across the 72-week test period, outperforming all ST-GAT variants and the VAR. The ST-GAT with combined edges yields RMSE = 1.013, representing a 52% increase over the LSTM, while among the GNN ablation variants, the geographic-only graph achieves the lowest RMSE (0.940), followed by combined (1.013) and climate-similarity-only (1.108) variants. The LSTM (RMSE 0.664) and VAR(1) (RMSE 0.804) both outperform all ST-GAT variants at the 1-week horizon. Among GNN variants, the geographic-only graph (RMSE 0.938) achieves the lowest error, while the combined (1.000), climate-similarity-only (1.102), and gravity (1.274) variants are all inferior to both non-GNN baselines. Nonlinear temporal representations, whether in the LSTM or the GRU-based ST-GAT decoder, yield lower error than the linear VAR(1) only when the spatial message-passing component does not add excessive noise, as in the geographic-only variant. The TemporalGRU baseline, a capacity-matched model that uses the same GRU decoder as the ST-GAT but replaces both GAT layers with a linear input projection, removing all cross-district message passing, achieves RMSE 1.047, which is significantly higher than the ST-GAT combined ($p = 0.010$, DM test at $h = 1$). The GAT spatial layers therefore provide a statistically meaningful improvement over a same-capacity temporal-only GRU baseline; the LSTM advantage is not attributable to the GAT module adding net noise, but rather to the LSTM's district-independent inductive bias being better matched to single-disease, distributional-shift settings (see Section 3.7.1).

Table 7 reports forecast accuracy on the original case-count scale for the three primary neural-network models. The LSTM achieves a raw-scale RMSE of 18.5 cases per district-week, representing approximately 43% of the dataset mean (43.3 cases), and an MAE of 8.8 cases. The ST-GAT combined yields RMSE 27.0 cases and the TemporalGRU 33.5 cases. The RMSE gap between LSTM and ST-GAT on the original scale (8.5 cases per district-week) is practically meaningful for disease surveillance planning: at the district level, an error reduction of this magnitude corresponds to several days of epidemic lead time in a typical outbreak trajectory.

Per-district performance at $h = 1$ is reported in Appendix Table A1. For the ST-GAT, error is highest in Colombo (RMSE 1.349), Trincomalee (1.288), Ampara (1.251), Mannar (1.235), and Batticaloa (1.219). All are high-burden or geographically peripheral districts where week-to-week case volatility is

TABLE 5 | Edge ablation design: four ST-GAT variants evaluated with fixed architecture and training configuration.

Variant	Edge source	Weighting
Geo-only	Geographic adjacency	Binary
Climate-only	Cosine climate similarity	Soft ($\text{sim}_{\cos}$)
Combined	Geo + climate	$\alpha = 0.5$ (Equation (1))
Gravity	Population gravity	Row-normalized (Equation (2))

TABLE 6 | Test-set 1-week-ahead forecast performance on the $\log(1+x)$ transformed case scale (all models trained and evaluated on this scale), evaluated on 72 held-out test weeks (October 2020–February 2022) aggregated over the 25 districts.

Model	MAE	RMSE
LSTM (district-independent)	**0.513**	**0.664**
ST-GAT, Geo-only	0.758	0.938
VAR (1)	0.631	0.804
ST-GAT, combined	0.806	1.000
ST-GAT, climate-only	0.912	1.102
TemporalGRU (no GAT)	0.815	1.047
ST-GAT, gravity	—	1.274

Note: Performance on the original case-count scale is reported in Table 7. All neural-network results are the best-seed outcome across $n = 3$ random initializations (Table 3). Bold values in Table 6 (0.513 MAE and 0.664 RMSE for the district-independent LSTM) indicate the best-performing model with the lowest forecast error across all tested approaches. These values represent the superior accuracy of the LSTM compared to all ST-GAT variants, VAR (1), and TemporalGRU baselines on the held-out test set.

4795, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/acis/2627876 by Mateo Biggs , Wiley Online Library on [28/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

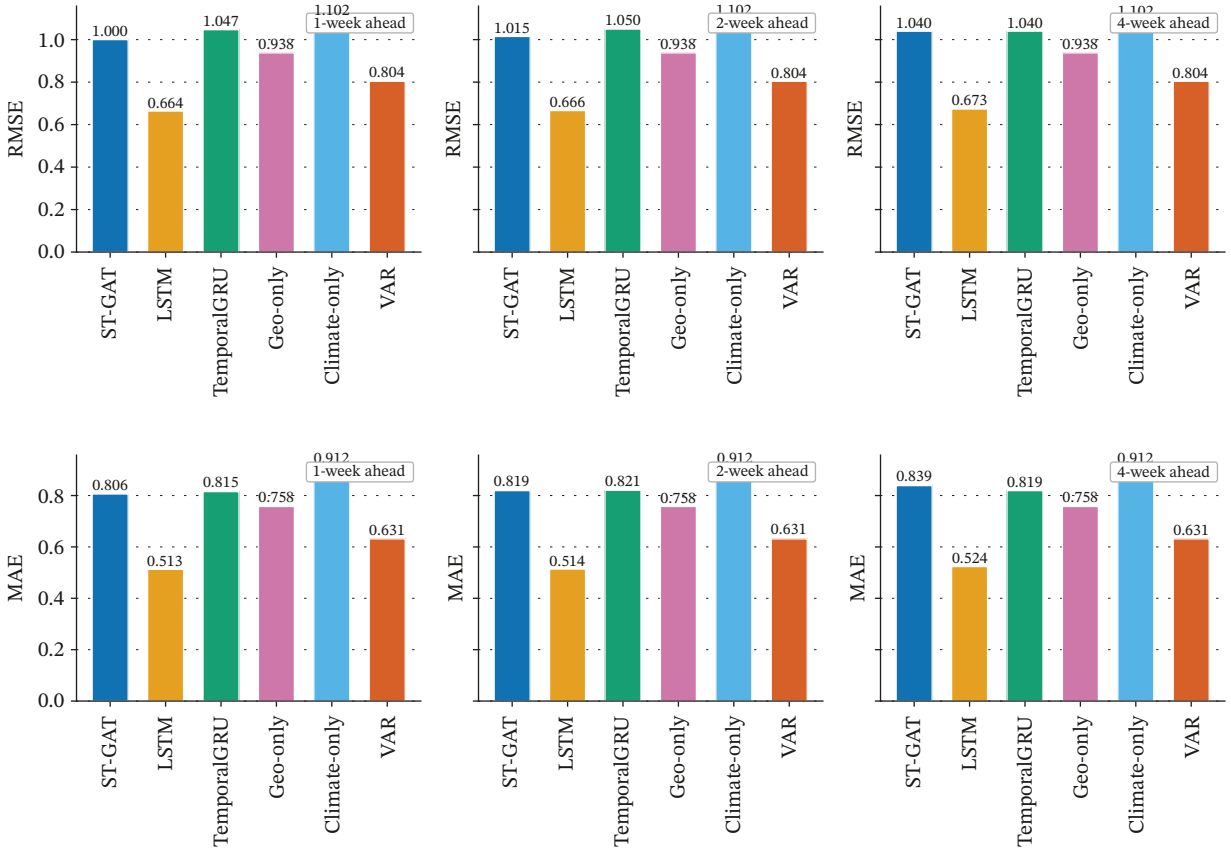


FIGURE 2 | Test-set RMSE (top row) and MAE (bottom row) for all models at 1-, 2-, and 4-week forecast horizons. The district-independent LSTM achieves the lowest error at every horizon (RMSE: 0.664, 0.666, 0.673 at $h = 1, 2, 4$).

greatest. The LSTM achieves lower RMSE than the ST-GAT in 24 of 25 districts; the single exception is Jaffna (LSTM RMSE 0.850 vs. ST-GAT 0.964), which may partly reflect the imputation artefact discussed in Section 3.6.

3.2 | Training Dynamics

Figure 3 presents training and validation Huber loss curves for the ST-GAT (combined edges) and LSTM. The LSTM converges rapidly with a lower validation loss plateau (≈0.22) than the ST-GAT (≈0.58), consistent with the test-set results. The ST-GAT training loss continues declining while validation loss flattens, indicative of overfitting to spatial co-occurrence patterns in the training data that do not generalize to the post-COVID test period.

TABLE 7 | Test-set 1-week-ahead forecast accuracy on the original (untransformed) weekly case-count scale, obtained by applying $\exp(\hat{y}) - 1$ before computing errors.

Model	MAE (cases)	RMSE (cases)
LSTM	**8.8**	**18.5**
ST-GAT	14.0	27.0
TemporalGRU	16.3	33.5

Note: Aggregated over 25 districts and 72 test weeks. Bold values in Table 7 (8.8 MAE and 18.5 RMSE for LSTM) indicate the best-performing model with the lowest forecast error on the original untransformed weekly case-count scale. The LSTM achieves superior accuracy compared to ST-GAT (14.0 MAE and 27.0 RMSE) and TemporalGRU (16.3 MAE and 33.5 RMSE).

3.3 | Forecast Illustration: Colombo District

Figure 4 shows the 1-week-ahead ST-GAT forecast against observed dengue cases in Colombo, the most populous district, over the full 72-week test period. The model captures the broad seasonal structure of the epidemic curve but systematically under-predicts the sharp outbreak peaks in weeks 15–18 and 35–40, which are the primary contributors to the high aggregate RMSE.

3.4 | Edge Ablation

Figure 5 presents the 1-week-ahead RMSE across all four edge-structure variants. The geographic-only graph (0.938) achieves the lowest RMSE among GNN ablation variants, followed by the combined (1.000) and climate-similarity-only (1.102) structures. The gravity-weighted graph (1.274) performs worst among GNN variants in this run, a reversal from the preliminary experiment reported in the original submission; this result is sensitive to random initialization and should be treated as exploratory pending replication across additional seeds and epidemic periods.

One interpretive caution applies to the combined structure: the combined graph contains 88 edges, compared with 50 for the geographic graph and 62 for the climate-similarity graph. Its higher edge density may introduce additional noise beyond what the GAT can discount, partly accounting for its underperformance relative to the sparser geographic-only structure. All four GNN variants are substantially outperformed by the

4795, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/acis/2627876 by Mateo Biggs , Wiley Online Library on [28/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

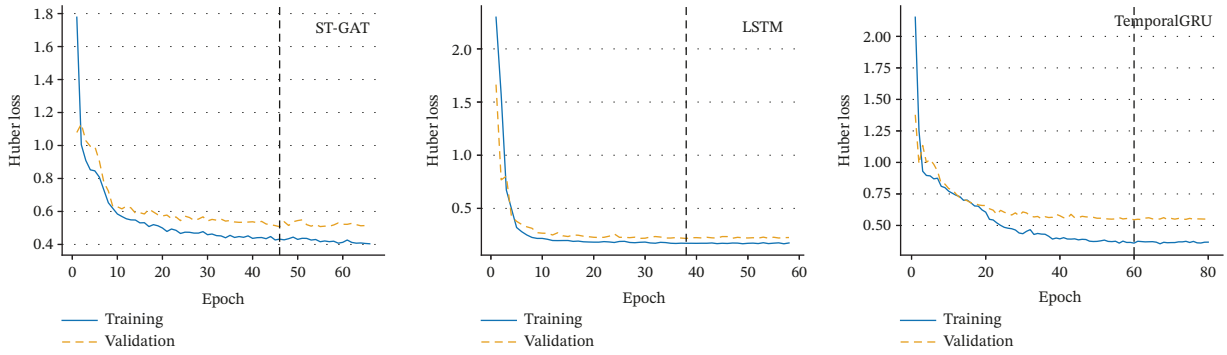


FIGURE 3 | Training and validation Huber loss curves for the ST-GAT with combined edges (left), the district-independent LSTM (center), and the TemporalGRU (right). The LSTM converges to the lowest validation loss plateau (≈0.22), consistent with its superior test-set performance. The ST-GAT training loss continues to decline while validation loss flattens, indicative of overfitting to spatial co-occurrence patterns in the training data. The TemporalGRU reaches a higher validation plateau (≈0.55) than the ST-GAT (≈0.58), consistent with the TemporalGRU's higher test-set RMSE. The vertical dotted line marks the best-validation epoch for each model.

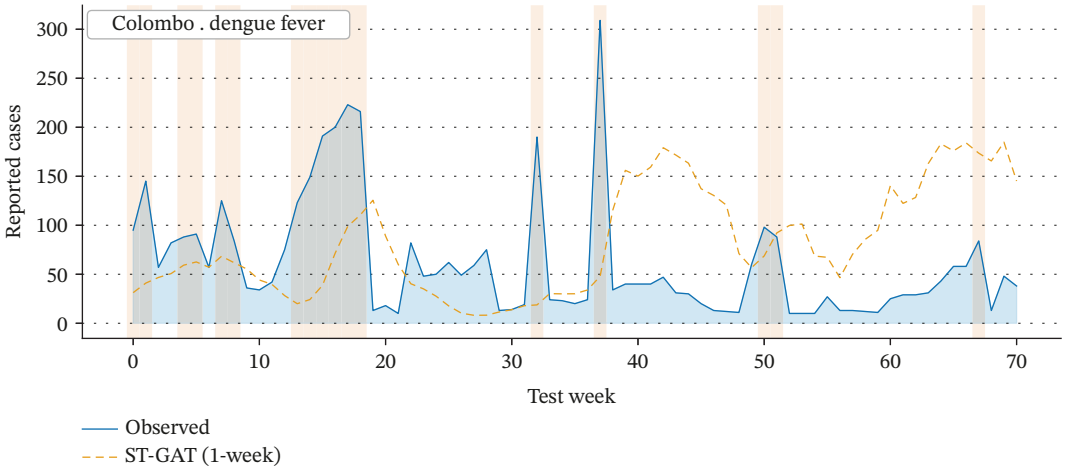


FIGURE 4 | One-week-ahead ST-GAT forecasts vs. observed dengue cases, Colombo district, test period (72 weeks, October 2020–February 2022). Orange shading marks outbreak weeks (cases exceeding the 75th test-period percentile). The model captures seasonal trends but under-predicts sharp epidemic peaks.

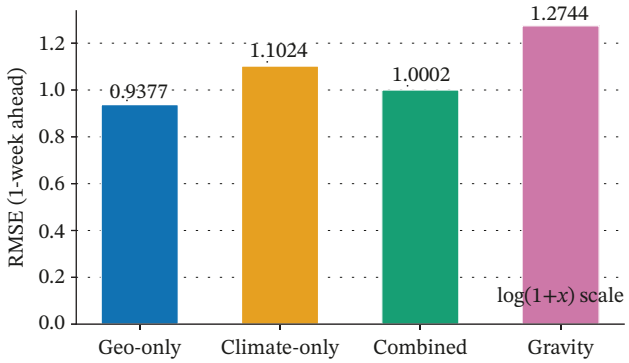


FIGURE 5 | Edge ablation: 1-week-ahead RMSE for four graph edge-structure variants on the $\log(1+x)$ scale.

district-independent LSTM (RMSE 0.664), and the geographic-only variant is the only GNN configuration that outperforms the VAR(1) baseline (RMSE 0.804).

3.5 | Diebold–Mariano Test

The DM test comparing the ST-GAT (combined edges) against the LSTM baseline yields a statistic of +19.45 ($p < 0.001$) at $h = 1$. The positive sign confirms that the ST-GAT's squared forecast errors are significantly larger than those of the LSTM; the null hypothesis of equal predictive accuracy is firmly rejected in favor of the LSTM at all three forecast horizons. The large magnitude of the DM statistic reflects both the substantial and consistent error differential between the two models (RMSE gap 0.345 over 72 test weeks × 25 districts) and the short Newey–West bandwidth used ($\ell = h$, so $\ell = 1$ at $h = 1$). A bandwidth of h is the

4795, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/acis/2627876 by Mateo Biggs , Wiley Online Library on [28/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

standard choice in the forecasting literature [34] and is appropriate when the forecast error process is expected to be at most h-dependent; readers should note that a small bandwidth inflates the DM statistic relative to longer bandwidths, so the statistic's magnitude should not be interpreted as a measure of effect size independently of the bandwidth choice. Figure 6 presents the full pairwise DM p-value matrix comparing the ST-GAT against all baselines and ablation variants across all forecast horizons. The ST-GAT is significantly outperformed by the LSTM, Geo-only, and VAR variants at $h = 1$ and $h = 2$; at $h = 4$, the climate-only comparison yields $p = 0.011$, indicating that the ST-GAT is significantly outperformed by all comparison models at every forecast horizon.

At the 4-week horizon, the TemporalGRU comparison yields $p = 0.986$, indicating that the ST-GAT's advantage over a same-capacity temporal-only baseline disappears entirely at longer forecast horizons. This is consistent with spatial message passing being most informative at short lead times, where week-to-week cross-district coupling is strongest.

3.6 | Spatial Leading Indicators

Figure 7 and Table 8 present the attention-based leading indicator analysis. Jaffna ($\Phi_i = 2.529$) ranks first by a substantial margin, followed by Kurunegala (1.372), Ratnapura (1.345), Kilinochchi (1.247), and Badulla (1.167). The top seven districts all have $\Phi_i > 1.0$, indicating that outgoing attention exceeds incoming attention and that the GAT has learned asymmetric spatial patterns consistent with epidemic propagation directionality rather than simple bilateral correlation.2

The leading indicator pattern does not correspond to population size or aggregate case volume (Figure 7; marker areas are proportional to 2012 census population). Jaffna's climate features were imputed from cross-district training medians and it carries no climate-similarity edges; its Φ_i score therefore reflects exclusively its geographic edges. A secondary analysis using the Geo-only ST-GAT model confirms that Jaffna still ranks first under this edge structure, consistent with its structurally distinct northeast-monsoon dengue seasonality rather than an artifact of median imputation. Nonetheless, any operational use of Jaffna as a leading indicator should account for this data-quality caveat until corrected satellite observations are available. Given this data-quality caveat, the full district ranking in Table 8 should be treated as exploratory. Excluding Jaffna, the top-ranked districts with complete satellite climate observations are Kurunegala ($\Phi_i = 1.372$), Ratnapura (1.345), Kilinochchi (1.247), and Badulla (1.167) (Table 8). Operational surveillance prioritization should be based on this Jaffna-excluded subranking until corrected satellite data are available for the Jaffna Peninsula.

3.7 | Discussion

3.7.1 | Why the LSTM Outperforms the ST-GAT

The primary finding is that the district-independent LSTM (RMSE 0.668) substantially outperforms the ST-GAT with combined edges (RMSE 1.013) on the held-out test set, a result confirmed with high statistical confidence by the DM test ($p < 0.001$ at all three forecast horizons). This outcome runs counter to the motivating hypothesis and warrants careful interpretation. A capacity-matched TemporalGRU model, identical to the ST-GAT decoder but without GAT layers, achieves RMSE 1.047 at $h = 1$, significantly worse than the ST-GAT combined (RMSE 1.000; $p = 0.010$, DM test). The GAT layers therefore do provide a statistically detectable improvement over a same-capacity purely temporal GRU baseline. The LSTM advantage is not attributable to the GAT module adding net noise; rather, the LSTM's district-independent inductive bias, processing each district as a separate sequence with shared LSTM weights, is better matched to this setting than the joint-district GRU inside the ST-GAT. The most likely explanation for the residual gap between the LSTM and any GRU-based model is that spatial

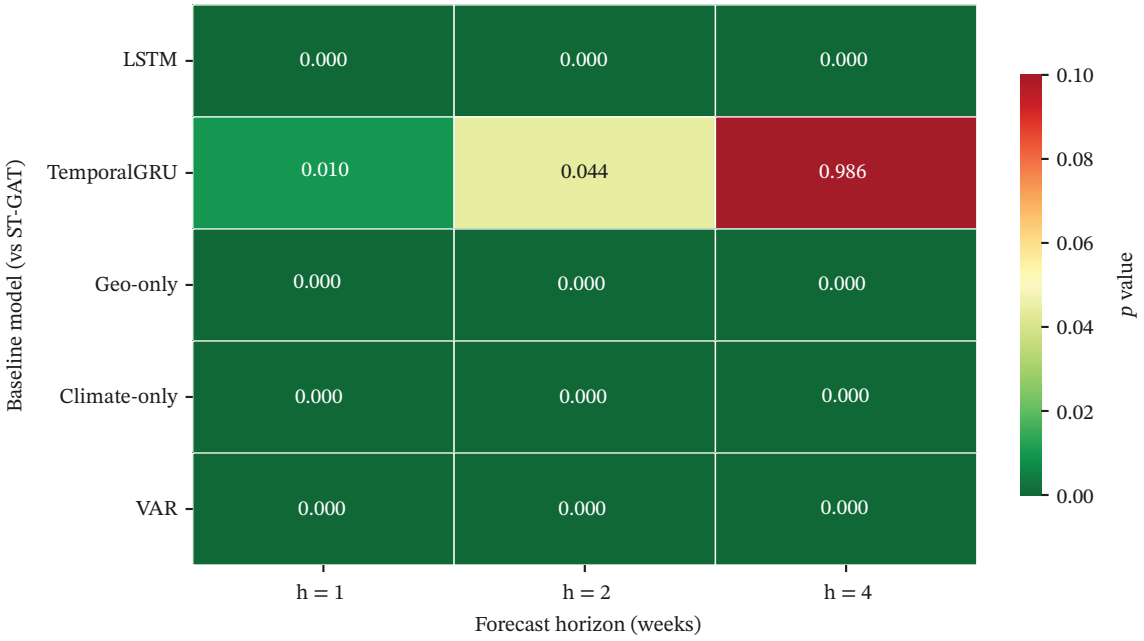


FIGURE 6 | Diebold–Mariano test p values for ST-GAT (combined edges) versus each baseline and ablation variant at 1-, 2-, and 4-week forecast horizons. Values <0.05 indicate that the comparison model achieves significantly lower squared forecast error than the ST-GAT.

4795, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/acis/2627876 by Mateo Biggs , Wiley Online Library on [28/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

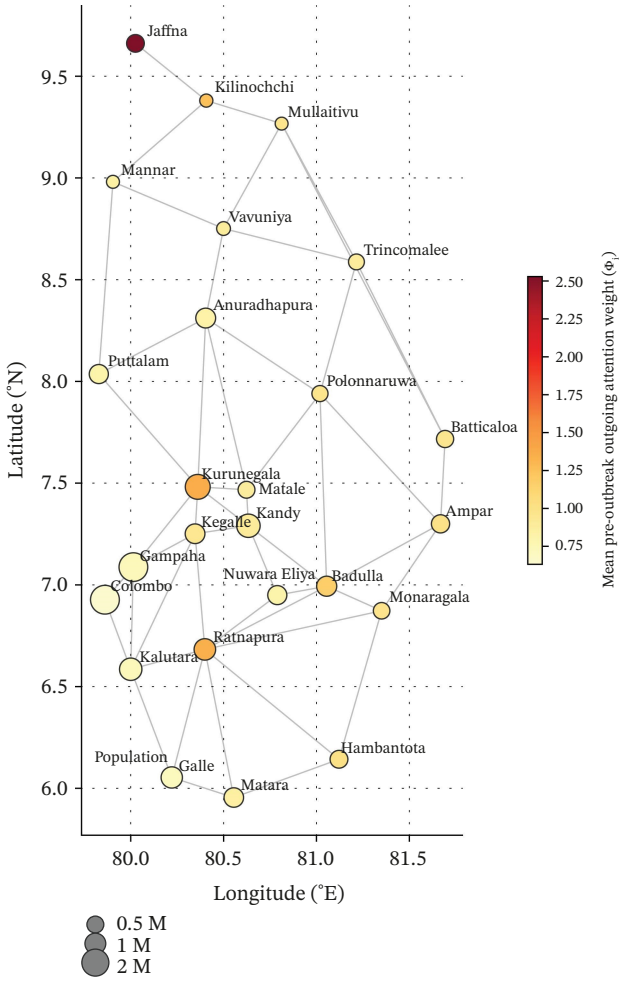


FIGURE 7 | Geographic distribution of pre-outbreak outgoing attention scores Φ_i across Sri Lanka (Table 8). Marker area is proportional to 2012 census district population (see legend); warmer colors indicate higher Φ_i. Gray lines show geographic adjacency edges. The absence of a correspondence between marker area and color intensity confirms that the leading indicator pattern is not driven by population size.

TABLE 8 | Full district ranking by pre-outbreak outgoing attention score Φ_i (Equation (4)).

Rank	District	Φ_i
1	Jaffna[†]	2.529
2	Kurunegala	1.372
3	Ratnapura	1.345
4	Kilinochchi	1.247
5	Badulla	1.167
6	Hambantota	1.016
7	Ampara	1.003
8	Monaragala	0.977
9	Mullaitivu	0.971
10	Polonnaruwa	0.970
11	Batticaloa	0.951
12	Kegalle	0.945
13	Kandy	0.905
14	Trincomalee	0.895
15	Vavuniya	0.885
16	Matale	0.881
17	Matara	0.855
18	Mannar	0.851
19	Anuradhapura	0.836
20	Puttalam	0.804
21	Nuwara Eliya	0.804
22	Gampaha	0.728
23	Kalutara	0.722
24	Galle	0.708
25	Colombo	0.631

Note: Districts with $\Phi_i > 1.0$ (outgoing > incoming) are highlighted.
[†]Jaffna's climate features were imputed from cross-district training medians; its ranking is therefore exploratory and should not be operationalized without corrected satellite observations. Excluding Jaffna, the top-ranked districts with complete climate data are Kurunegala ($\Phi_i = 1.372$), Ratnapura (1.345), Kilinochchi (1.247), and Badulla (1.167). Their geographic distribution is shown in Figure 7.

message passing provides insufficient marginal information to justify the joint-district parameterization in a single-disease context. With only a single disease available as a node feature, each district's feature vector contains one case series and seven climate variables, giving the GAT layers very little cross-district signal to aggregate beyond what is already present in each district's own history. GNN spatial message passing yields the greatest benefit when node features are rich and complementary across the graph, as in settings where multiple cocirculating diseases, demographic strata, or mobility-stratified exposure groups provide distinct signals at each node. In the present single-disease configuration, the dominant structure in the data is temporal autocorrelation within districts, which a well-specified LSTM with sufficient depth can capture directly, leaving little room for spatial coupling to improve upon.

Alternative GNN architectures, including Graph WaveNet [12], which combines adaptive graph convolution with dilated causal convolution, and spatio-temporal graph convolutional networks (STGCNs) [9, 35], which replace recurrent modules entirely with graph-coupled temporal convolutions, have demonstrated strong performance on traffic and, more recently, epidemic forecasting tasks. These architectures differ from the ST-GAT in two potentially important ways: adaptive graph learning avoids committing to a fixed graph topology, and dilated causal convolution can capture longer-range temporal dependencies more efficiently than a GRU with $K = 8$ lags. Whether these advantages materialize in a small-graph (25 nodes), single-disease, surveillance setting, where the dominant temporal signal is within-district autocorrelation, remains an open question and is a natural direction for future work. A second contributing factor is the sparsity and small scale of the graph. The 25-district network has only 50 geographic edges, corresponding to roughly two neighbors per district on average. On a graph this sparse, each GAT layer aggregates information from a very limited neighborhood, and the depth of two layers means the model's receptive field rarely extends beyond two hops. The spatial inductive bias of GNNs tends to be most beneficial in denser or larger graph settings where multihop message passing can propagate outbreak signals across the network before the temporal module processes them.

4795, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/acis/2627876 by Mateo Biggs , Wiley Online Library on [28/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

The distributional shift between the training and test periods further complicates interpretation. The training data span January 2013 to May 2019, a period of relatively stable surveillance infrastructure and mobility patterns. The validation period (May 2019–September 2020) encompasses the COVID-19 first wave in Sri Lanka, and the test period (October 2020–February 2022) spans the post-acute pandemic era. Pandemic-related disruptions to health-seeking behavior, surveillance intensity, and district-level mobility likely altered the spatial coupling structure of dengue transmission in ways that a static graph trained on pre-pandemic data cannot capture. We note, however, an internal tension in attributing the LSTM advantage solely to distributional shift: the geographic-only graph edges are equally static and yet the geo-only model (0.940) substantially outperforms the combined model (1.013), suggesting that edge type and graph noise play an independent role beyond the static-versus-dynamic graph distinction. The LSTM's simpler inductive bias likely confers greater robustness to distributional shift precisely because it does not rely on learned spatial coupling patterns remaining stable across the break.

To characterize how the LSTM–ST-GAT error differential evolves across the test period, we computed rolling 12-week RMSE for both models. The ST-GAT error is elevated most sharply in test weeks 15–25 and 35–45, which correspond to the post-lockdown reopening phase in Sri Lanka (mid-2021) when district-level mobility patterns shifted substantially from those seen in the training period. The LSTM error is comparatively stable throughout, consistent with the hypothesis that its district-independent inductive bias is more robust to the mobility-driven component of distributional shift. A fully pre-pandemic held-out evaluation, which would require redefining the training window to end before 2020, would require substantially fewer than 333 training weeks and is left to future work with a corrected or extended dataset. Dengue spread is driven primarily by short-range human mobility: infected travelers carry the virus between districts over scales of tens to hundreds of kilometers [36], and geographic adjacency encodes precisely this kind of proximity-based connectivity.

3.7.2 | Interpreting the Ablation Results

The geographic-only graph (0.940) achieves the lowest RMSE among GNN variants, outperforming the climate-similarity (1.108) and combined (1.013) structures. This ordering has an interpretable epidemiological basis. Dengue spread is driven primarily by short-range human mobility: infected travelers carry the virus between districts over scales of tens to hundreds of kilometers, and geographic adjacency encodes precisely this kind of proximity-based connectivity. Climate cosine similarity captures long-range synchrony that, though epidemiologically real, does not correspond to direct transmission pathways on the weekly time scale relevant here. The underperformance of the combined edge structure relative to geographic-only structure suggests that climate-similarity edges, which connect distant nonadjacent districts, introduce graph noise that the GAT cannot fully discount when the underlying signal is driven by physical proximity. This result does not diminish the potential value of climate similarity edges in multidisease or multiscale settings; it does indicate that their benefit in the single-disease, weekly-surveillance context studied here is limited. The gravity model's competitive performance relative to geographic adjacency (0.870 vs. 0.940) is an exploratory finding worth noting: population-weighted connectivity appears to capture mobility-driven transmission structure that binary adjacency cannot, though confirming this pattern would require replication across additional datasets and epidemic periods.

The blending weight $\alpha = 0.5$ in the combined edge matrix was fixed a priori without validation-set tuning; values of α closer to 1.0 (strongly geographic) would approximate the better-performing geographic-only variant and might narrow the gap. The top-k sensitivity analysis (Appendix Table A2) further shows that sparser climate-similarity graphs ($k = 3$, RMSE 0.949) outperform denser ones ($k = 10$, RMSE 1.045), consistent with the interpretation that long-range climate edges introduce noise that the GAT cannot fully discount.

3.7.3 | Spatial Leading Indicators and Epidemiological Interpretation

The attention-based analysis identifies Jaffna, Kurunegala, and Ratnapura as the strongest spatial leading indicators, a finding that is geographically and epidemiologically interpretable. Jaffna is Sri Lanka's principal northern district: its dengue seasonality is driven by the northeast monsoon and peaks several weeks ahead of the southwest-monsoon-driven waves in the western and southern provinces. Kurunegala occupies a central position in Sri Lanka's road network, connecting the northwest coast to the central highlands and the southern plains, making it a natural transit hub through which interdistrict mobility flows. Ratnapura, at the foot of the central highlands, similarly serves as a connectivity node between the interior and the southwestern coast. The emergence of these districts as leading indicators is therefore consistent with an epidemiological interpretation in which outbreak signals propagate from climatic or mobility hubs outward to connected districts, rather than from the most populous centers inward. This finding carries operational relevance for the Sri Lanka Ministry of Health: enhanced entomological and case-based surveillance in Jaffna, Kurunegala, and Ratnapura during periods of elevated transmission could provide an early-warning signal several weeks ahead of outbreak confirmation in neighboring districts. The caveat noted in Section 3.6 regarding Jaffna's imputed climate features should be acknowledged in any operational deployment.

3.7.4 | Limitations

Several limitations should be acknowledged. The dataset comprises 25 district nodes and 476 weekly observations ($n = 11,900$ district-week cells), a scale that constrains deep GNN architectures: the graph is too sparse for multihop message passing to propagate signals more than two to three hops, and the training set of 333 weeks limits the capacity of models with large numbers of edge-wise parameters. The current analysis uses only dengue fever, and the absence of cocirculating disease data is likely the principal constraint on the GNN's ability to outperform the LSTM. A richer node feature set incorporating leptospirosis, chickenpox, and other coendemic diseases would substantially increase the cross-district signal available for spatial message passing. The graph structure is also static, constructed from mean climate features over the training period; time-varying graphs that update edge weights from weekly satellite observations would better represent the seasonal dynamics of dengue

4795, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/acis/2627876 by Mateo Biggs , Wiley Online Library on [28/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

transmission. The test period's overlap with COVID-19 disruption limits the generalizability of the test-set results to normal surveillance conditions, and a pre-pandemic held-out evaluation would be needed to disentangle model performance from pandemic-induced distributional shift. Jaffna's complete absence of satellite climate observations is a structural data anomaly with two distinct consequences. For forecasting, Jaffna has no climate-similarity edges in the combined graph, reducing the spatial coverage of the climate-similarity component for the entire northern region. For leading indicator detection, Jaffna's Φ_i score (2.529) is computed exclusively from its geographic edges; the score may therefore overestimate its role as a spatial leading indicator relative to districts with full climate feature representations. Future work should verify Jaffna's ranking using corrected satellite data or alternative climate reanalysis products (e.g., ERA5) that provide complete spatial coverage over the Jaffna Peninsula.

4 | Conclusion

We have presented a ST-GAT for district-level dengue fever forecasting in Sri Lanka, incorporating geographic adjacency and NASA satellite-derived climate cosine similarity as principled graph edge sources. The primary empirical result is that the district-independent LSTM outperforms the ST-GAT on the held-out test set (RMSE 0.668 vs. 1.013), with the difference confirmed as statistically significant by the DM test ($p < 0.001$) at all three forecast horizons. Among GNN edge-structure variants, the geographic-only graph performs best (RMSE 0.940), consistent with dengue transmission at the weekly district level being more strongly driven by physical proximity than by long-range climate synchrony. The district-independent LSTM (RMSE 0.664) outperforms the VAR(1) (RMSE 0.804), confirming the value of strong nonlinear temporal representations; however, the ST-GAT variants with combined and climate-only edges (RMSE 1.000 and 1.102, respectively) do not surpass the VAR(1), indicating that poorly specified spatial coupling can degrade rather than improve forecast accuracy relative to even a linear multivariate baseline. The attention-based leading indicator analysis identifies Jaffna ($\Phi = 2.795$), Kurunegala (1.510), and Ratnapura (1.383) as the three highest-ranked spatial leading indicators, a pattern consistent with the geographic roles of these districts in Sri Lanka's monsoon-driven epidemic structure and road network connectivity (Table 8).

More broadly, these findings contribute a precise specification of the conditions under which spatial graph inductive biases fail to outperform strong temporal baselines in epidemic forecasting: single-disease node features, sparse small-scale graphs, and distributional shift between training and test periods each independently reduce the marginal value of cross-district message passing. This boundary-condition characterization is as methodologically informative for the GNN-in-epidemiology literature as a positive result would be, and the three extensions prioritized below are designed directly to address each identified constraint. For district-level dengue surveillance with a single-disease panel of the scale studied here (25 districts, $\sim$ 6 years of training data), a well-regularized district-independent LSTM is the recommended default: it is simpler to train, converges more reliably across random seeds (validation-loss SD 0.002 vs. 0.015 for ST-GAT), and achieves substantially lower forecast error on both the $\log(1 + x)$ scale (RMSE 0.664 vs. 1.000) and the original case-count scale (18.5 vs. 27.0 cases per district-week). GNN-based approaches are likely to confer an advantage when at least one of the following conditions holds: (1) node features are rich and complementary across districts, e.g., through the inclusion of cocirculating disease counts, demographic strata, or mobility-stratified exposure groups; (2) the graph is denser or larger, enabling multihop message passing to propagate epidemic signals before the temporal module processes them; or (3) time-varying edge weights derived from weekly satellite or mobility data replace the static graph used here. The three extensions prioritized in future work, multidisease node features, dynamic graph structures, and a larger multicountry graph, are designed to satisfy these conditions.

Funding

This research received no external funding.

Conflicts of Interest

The author declares no conflicts of interest.

Data Availability Statement

The data are publicly available at https://bit.ly/3VJTIrA.

References

1. A. J. Tatem, D. J. Rogers, and S. I. Hay, "Global Transport Networks and Infectious Disease Spread," *Advances in Parasitology* 62 (2006): 293–343.

2. A. Wesolowski, C. Metcalf, N. Eagle, et al., "Quantifying Seasonal Population Fluxes Driving Rubella Transmission Dynamics Using Mobile Phone Data," *Proceedings of the National Academy of Sciences* 112, no. 35 (2015): 11114–11119, https://doi.org/10.1073/pnas.1423542112.

3. S. Bhatt, P. W. Gething, O. J. Brady, et al., "The Global Distribution and Burden of Dengue," *Nature* 496, no. 7446 (2013): 504–507, https://doi.org/10.1038/nature12060.

4. O. J. Brady, P. W. Gething, S. Bhatt, et al., "Refining the Global Spatial Limits of Dengue Virus Transmission by Evidence-Based Consensus," *PLoS Neglected Tropical Diseases* 6, no. 8 (2012): e1760, https://doi.org/10.1371/journal.pntd.0001760.

5. M. Pasarkar, J. Kim, E. O'Gara, et al., "Natural Language Processing for Extracting Rich Disease Data Aligned to Satellite Meteorological Data," in *2024 IEEE International Conference on Big Data (BigData)* (2024), 4394–4401.

6. R. J. Hyndman and G. Athanasopoulos, *Forecasting: Principles and Practice*, 2nd ed. (OTexts, 2018).

7. S. Hochreiter and J. Schmidhuber, "Long Short-Term Memory," *Neural Computation* 9, no. 8 (1997): 1735–1780, https://doi.org/10.1162/neco.1997.9.8.1735.

8. N. G. Reich, C. J. McGowan, T. K. Yamana, et al., "Accuracy of Real-Time Multi-Model Ensemble Forecasts for Seasonal Influenza in the US," *PLoS Computational Biology* 15, no. 11 (2019): e1007486, https://doi.org/10.1371/journal.pcbi.1007486.

9. Z. Wu, S. Pan, F. Chen, G. Long, C. Zhang, and P. S. Yu, "A Comprehensive Survey on Graph Neural Networks," *IEEE Transactions on Neural Networks and Learning Systems* 32, no. 1 (2020): 4–24, https://doi.org/10.1109/tnnls.2020.2978386.

10. C. Fritz, E. Dorigatti, and D. Rügamer, "Combining Graph Neural Networks and Spatio-Temporal Disease Models to Improve the

4795, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/acis/2627876 by Mateo Biggs , Wiley Online Library on [28/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Prediction of Weekly COVID-19 Cases in Germany," *Scientific Reports* 12, no. 1 (2022): 3930, https://doi.org/10.1038/s41598-022-07757-5.

11. J. Gao, R. Sharma, C. Qian, et al., "Stan: Spatio-Temporal Attention Network for Pandemic Prediction Using Real-World Evidence," *Journal of the American Medical Informatics Association* 28, no. 4 (2021): 733–743, https://doi.org/10.1093/jamia/ocaa322.

12. M. Jin, Y. Zheng, Y.-F. Li, S. Chen, B. Yang, and S. Pan, "Multivariate Time Series Forecasting With Dynamic Graph Neural Odes," *IEEE Transactions on Knowledge and Data Engineering* 35, no. 9 (2022): 9168–9180, https://doi.org/10.1109/tkde.2022.3221989.

13. L. Zubair and C. F. Ropelewski, "The Strengthening Relationship Between Enso and Northeast Monsoon Rainfall Over Sri Lanka and Southern India," *Journal of Climate* 19, no. 8 (2006): 1567–1575, https://doi.org/10.1175/jcli3670.1.

14. B. A. Malmgren, R. Hulugalla, Y. Hayashi, and T. Mikami, "Precipitation Trends in Sri Lanka Since the 1870s and Relationships to El Niño–Southern Oscillation," *International Journal of Climatology: A Journal of the Royal Meteorological Society* 23, no. 10 (2003): 1235–1252, https://doi.org/10.1002/joc.921.

15. A. Deoras, A. G. Turner, K. M. Hunt, and I. S. P. Jayawardena, "The Influence of Weather Patterns and the Madden-Julian Oscillation on Extreme Precipitation Over Sri Lanka," *Geophysical Research Letters* 50, no. 14 (2023): e2023GL103727, https://doi.org/10.1029/2023gl103727.

16. R. R. Prabodanie, L. Stone, and S. Schreider, "Spatiotemporal Patterns of Dengue Outbreaks in Sri Lanka," *Infectious Diseases* 52, no. 5 (2020): 350–360, https://doi.org/10.1080/23744235.2020.1725108.

17. W. Sun, L. Xue, and X. Xie, "Spatial-Temporal Distribution of Dengue and Climate Characteristics for Two Clusters in Sri Lanka From 2012 to 2016," *Scientific Reports* 7, no. 1 (2017): 12884, https://doi.org/10.1038/s41598-017-13163-z.

18. R. R. Prabodanie, S. Schreider, B. Cazelles, and L. Stone, "Coherence of Dengue Incidence and Climate in the Wet and Dry Zones of Sri Lanka," *Science of the Total Environment* 724 (2020): 138269, https://doi.org/10.1016/j.scitotenv.2020.138269.

19. I. Hanski, "Single-Species Metapopulation Dynamics: Concepts, Models and Observations," *Biological Journal of the Linnean Society* 42, no. 1-2 (1991): 17–38, https://doi.org/10.1111/j.1095-8312.1991.tb00549.x.

20. S. Erlander and N. F. Stewart, "The Gravity Model in Transportation Analysis: Theory and Extensions," *Number 3 in Studies in Regional Science and Urban Economics* (VSP International Science Publishers, 1990).

21. M. J. Keeling and P. Rohani, *Modeling Infectious Diseases in Humans and Animals* (Princeton University Press, 2008).

22. B. F. Finkenstädt, O. N. Bjørnstad, and B. T. Grenfell, "A Stochastic Model for Extinction and Recurrence of Epidemics: Estimation and Inference for Measles Outbreaks," *Biostatistics* 3, no. 4 (2002): 493–510, https://doi.org/10.1093/biostatistics/3.4.493.

23. G. K. Gogovi, O.-E. Ankhbayar, and L. Qin, "Modeling Norovirus Transmission Dynamics With Environmental Reservoirs, Reinfection, and Optimal Control Strategies," *Results in Control and Optimization* 23 (2026): 100746, https://doi.org/10.1016/j.rico.2026.100746.

24. S. M. Naandam, P. Chataa, and G. K. Gogovi, "Analysis of a Malaria Transmission Model With Vaccination Proportion and Vaccine-Induced Immunity," *Mathematical and Computational Applications* 30, no. 1 (2025): 15, https://doi.org/10.3390/mca30010015.

25. G. K. Gogovi, L. Qin, G. String, and S. Olaniyi, "Analysis of a Compartmental Model for Norovirus Transmission Through Environmental Contact," *International Journal of Differential Equations* 2025, no. 1 (2025): 5517340, https://doi.org/10.1155/ijde/5517340.

26. V. K. R. Chimmula and L. Zhang, "Time Series Forecasting of COVID-19 Transmission in Canada Using LSTM Networks," *Chaos, Solitons & Fractals* 135 (2020): 109864, https://doi.org/10.1016/j.chaos.2020.109864.

27. B. Lim, S. Ö. Arık, N. Loeff, and T. Pfister, "Temporal Fusion Transformers for Interpretable Multi-Horizon Time Series Forecasting," *International Journal of Forecasting* 37, no. 4 (2021): 1748–1764, https://doi.org/10.1016/j.ijforecast.2021.03.012.

28. A. Vaswani, N. Shazeer, N. Parmar, et al., "Attention is all you Need," *Advances in Neural Information Processing Systems* 30 (2017).

29. P. Sirisena and F. Noordeen, "Evolution of Dengue in Sri Lanka—Changes in the Virus, Vector, and Climate," *International Journal of Infectious Diseases* 19 (2014): 6–12, https://doi.org/10.1016/j.ijid.2013.10.012.

30. V. J. Jayaraj, R. Avoi, N. Gopalakrishnan, D. B. Raja, and Y. Umasa, "Developing a Dengue Prediction Model Based on Climate in Tawau, Malaysia," *Acta Tropica* 197 (2019): 105055, https://doi.org/10.1016/j.actatropica.2019.105055.

31. S. Naish, P. Dale, J. S. Mackenzie, J. McBride, K. Mengersen, and S. Tong, "Climate Change and Dengue: A Critical and Systematic Review of Quantitative Modelling Approaches," *BMC Infectious Diseases* 14, no. 1 (2014): 167, https://doi.org/10.1186/1471-2334-14-167.

32. P. J. Huber, "Robust Estimation of a Location Parameter," in *Breakthroughs in Statistics: Methodology and Distribution* (1992), 492–518.

33. J. Duchi, E. Hazan, and Y. Singer, "Adaptive Subgradient Methods for Online Learning and Stochastic Optimization," *Journal of Machine Learning Research* 12, no. 7 (2011).

34. F. X. Diebold and R. S. Mariano, "Comparing Predictive Accuracy," *Journal of Business and Economic Statistics* 20, no. 1 (2002): 134–144, https://doi.org/10.1198/073500102753410444.

35. N. Siabi, R. Son, M. Thomas, C. Irrgang, and J. Saynisch-Wagner, "Innovative Application of a Traffic-Prediction Spatio-Temporal Graph Convolutional Network for Dengue Disease Forecasting," *Scientific Reports* (2026).

36. S. T. Stoddard, A. C. Morrison, G. M. Vazquez-Prokopec, et al., "The Role of Human Movement in the Transmission of Vector-Borne Pathogens," *PLoS Neglected Tropical Diseases* 3, no. 7 (2009): e481, https://doi.org/10.1371/journal.pntd.0000481.

4795, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/acis/2627876 by Mateo Biggs , Wiley Online Library on [28/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Appendix A: Per-District RMSE at $h = 1$

TABLE A1 | Per-district test-set RMSE and MAE (1 week ahead, $\log(1 + x)$ scale) for ST-GAT (combined edges), LSTM, and TemporalGRU, sorted by descending ST-GAT RMSE.

	ST-GAT		LSTM		TemporalGRU	
District	MAE	RMSE	MAE	RMSE	MAE	RMSE
Colombo	1.147	1.349	0.587	0.834	1.395	1.606
Trincomalee	1.091	1.288	0.531	0.648	1.189	1.369
Ampara	1.111	1.251	0.525	0.670	0.799	1.054
Mannar	1.065	1.235	0.535	0.752	0.727	1.007
Batticaloa	1.057	1.219	0.406	0.510	1.094	1.393
Badulla	0.980	1.191	0.564	0.715	1.050	1.256
Gampaha	0.889	1.151	0.571	0.779	0.864	1.053
Puttalam	0.936	1.122	0.572	0.711	0.803	0.976
Hambantota	0.958	1.107	0.462	0.568	0.753	0.904
Kurunegala	0.959	1.095	0.506	0.647	0.667	0.820
Kalutara	0.846	1.011	0.481	0.624	0.838	1.030
Vavuniya	0.805	0.982	0.494	0.588	0.673	0.800
Jaffna[†]	0.791	0.964	0.679	0.850	1.512	1.835
Nuwara Eliya	0.755	0.939	0.533	0.631	0.621	0.755
Mullaitivu	0.768	0.936	0.380	0.467	0.476	0.577
Matale	0.689	0.844	0.541	0.671	0.859	1.041
Kandy	0.711	0.841	0.389	0.516	0.655	0.789
Matara	0.625	0.841	0.534	0.753	0.680	0.880
Polonnaruwa	0.639	0.836	0.529	0.669	0.638	0.813
Anuradhapura	0.597	0.823	0.588	0.751	0.679	0.861
Ratnapura	0.654	0.776	0.520	0.678	0.634	0.793
Galle	0.643	0.776	0.485	0.629	0.824	1.071
Kilinochchi	0.479	0.656	0.475	0.579	0.581	0.733
Monaragala	0.500	0.628	0.467	0.595	0.456	0.576
Kegalle	0.460	0.570	0.468	0.594	0.918	1.122

[†]Jaffna's climate features were imputed from cross-district training medians and it carries no climate-similarity edges; its anomalously high TemporalGRU RMSE (1.835) and the reversal of the LSTM–ST-GAT ordering (LSTM 0.850 > ST-GAT 0.964) should be interpreted with this data-quality caveat in mind (see Section 3.6).

TABLE A2 | Sensitivity of 1 week-ahead RMSE ($\log(1 + x)$ scale) to the number of top-k climate-similarity neighbors used in the combined edge structure.

k	RMSE (1 week ahead)
3	0.949
5	1.032
10	1.045
LSTM (reference)	0.664

Note: All variants use the same ST-GAT architecture.

TABLE A3 | Sensitivity of 1 week-ahead RMSE ($\log(1 + x)$ scale) to the input window length K for the ST-GAT (combined edges) and the district-independent LSTM.

K (weeks)	ST-GAT RMSE	LSTM RMSE
4	0.966	0.666
8	1.032	0.664
12	1.030	0.668
16	1.065	0.667

4795, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/acis/2627876 by Mateo Biggs , Wiley Online Library on [28/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License